

Data for engineering materials

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This appendix lists the names and typical applications of common engineering materials, together with data for their properties.

A1 Names and applications: metals and alloys

Materials		Applications
Ferrous	Cast irons	Automotive parts, engine blocks, machine tool structural parts, lathe beds
	High carbon steels	Cutting tools, springs, bearings, cranks, shafts, railway track
	Medium carbon steels	General mechanical engineering (tools, bearings, gears, shafts, bearings)
	Low carbon steels	Steel structures ('mild steel')—bridges, oil rigs, ships; reinforcement for concrete; automotive parts, car body panels; galvanised sheet; packaging (cans, drums)
	Low alloy steels	Springs, tools, ball bearings, automotive parts (gears, connecting rods, etc.)
	Stainless steels	Transport, chemical and food processing plant, nuclear plant, domestic ware (cutlery, washing machines, stoves), surgical implements, pipes, pressure vessels, liquid gas containers
Non-ferrous	Aluminum alloys	
	Casting alloys	Automotive parts (cylinder blocks), domestic appliances (irons)
	Non-heat-treatable alloys	Electrical conductors, heat exchangers, foil, tubes, saucepans, beverage cans, lightweight ships, architectural panels
	Heat-treatable alloys	Aerospace engineering, automotive bodies and panels, lightweight structures and ships
	Copper alloys	Electrical conductors and wire, electronic circuit boards, heat exchangers, boilers, cookware, coinage, sculptures
	Lead alloys	Roof and wall cladding, solder, X-ray shielding, battery electrodes
	Magnesium alloys	Automotive castings, wheels, general lightweight castings for transport, nuclear fuel containers; principal alloying addition to aluminum alloys
	Nickel alloys	Gas turbines and jet engines, thermocouples, coinage; alloying addition to austenitic stainless steels
	Titanium alloys	Aircraft turbine blades; general structural aerospace applications; biomedical implants
	Zinc alloys	Die castings (automotive, domestic appliances, toys, handles); coating on galvanised steel

A2 Names and applications: polymers and foams

Materials		Abbreviation	Applications
Elastomer	Butyl rubber		Tyres, seals, anti-vibration mountings, electrical insulation, tubing
	Ethylene-vinyl-acetate	EVA	Bags, films, packaging, gloves, insulation, running shoes
	Isoprene	IR	Tyres, inner tubes, insulation, tubing, shoes
	Natural rubber	NR	Gloves, tyres, electrical insulation, tubing
	Polychloroprene (neoprene)	CR	Wetsuits, O-rings and seals, footwear
	Polyurethane elastomers	el-PU	Packaging, hoses, adhesives, fabric coating
	Silicone elastomers		Electrical insulation, electronic encapsulation, medical implants
Thermoplastic	Acrylonitrile butadiene styrene	ABS	Communication appliances, automotive interiors, luggage, toys, boats
	Cellulose polymers	CA	Tool and cutlery handles, decorative trim, pens
	Ionomer	I	Packaging, golf balls, blister packs, bottles
	Polyamides (nylons)	PA	Gears, bearings; plumbing, packaging, bottles, fabrics, textiles, ropes
	Polycarbonate	PC	Safety goggles, shields, helmets; light fittings, medical components
	Polyetheretherketone	PEEK	Electrical connectors, racing car parts, fibre composites
	Polyethylene	PE	Packaging, bags, squeeze tubes, toys, artificial joints
	Polyethylene terephthalate	PET	Blow moulded bottles, film, audio/video tape, sails
	Polymethyl methacrylate (acrylic)	PMMA	Aircraft windows, lenses, reflectors, lights, compact discs
	Polyoxymethylene (acetal)	POM	Zips, domestic and appliance parts, handles
	Polypropylene	PP	Ropes, garden furniture, pipes, kettles, electrical insulation, Astroturf
	Polystyrene	PS	Toys, packaging, cutlery, audio cassette/CD cases
	Polyurethane thermoplastics	tp-PU	Cushioning, seating, shoe soles, hoses, car bumpers, insulation
	Polyvinylchloride	PVC	Pipes, gutters, window frames, packaging
Thermoset	Polytetrafluoroethylene (Teflon)	PTFE	Nonstick coatings, bearings, skis, electrical insulation, tape
	Epoxies	EP	Adhesives, fibre composites, electronic encapsulation
	Phenolics	PHEN	Electrical plugs, sockets, cookware, handles, adhesives
	Polyester	PEST	Furniture, boats, sports goods
Polymer foams	Flexible polymer foam		Packaging, buoyancy, cushioning, sponges, sleeping mats
	Rigid polymer foam		Thermal insulation, sandwich panels, packaging, buoyancy

A3 Names and applications: composites, glasses, ceramics, and natural materials

Materials		Applications
Composites		
Metal	Aluminum/silicon carbide	Automotive parts, sports goods
Polymer	CFRP	Lightweight structural parts (aerospace, bike frames, sports goods, boat hulls and oars, springs)
	GFRP	Boat hulls, automotive parts, chemical plant
Glasses	Borosilicate glass	Ovenware, laboratory ware, headlights
	Glass ceramic	Cookware, lasers, telescope mirrors
	Silica glass	High-performance windows, crucibles, high-temperature applications
	Soda-lime glass	Windows, bottles, tubing, light bulbs, pottery glazes
Ceramics		
Porous	Brick	Buildings
	Concrete	General civil engineering construction
	Stone	Buildings, architecture, sculpture
Technical	Alumina	Cutting tools, spark plugs, microcircuit substrates, valves
	Aluminum nitride	Microcircuit substrates and heat sinks
	Boron carbide	Lightweight armour, nozzles, dies, precision tool parts
	Silicon	Microcircuits, semiconductors, precision instruments, IR windows, MEMS
	Silicon carbide	High-temperature equipment, abrasive polishing grits, bearings, armour
	Silicon nitride	Bearings, cutting tools, dies, engine parts
	Tungsten carbide	Cutting tools, drills, abrasives
Natural		
	Bamboo	Building, scaffolding, paper, ropes, baskets, furniture
	Cork	Corks and bungs, seals, floats, packaging, flooring
	Leather	Shoes, clothing, bags, drive-belts
	Wood	Construction, flooring, doors, furniture, packaging, sports goods

A4 Melting temperature, T_m , and glass temperature T_g ¹

T_m or T_g (°C)		
Metals		
Ferrous	Cast irons	1130–1250
	High carbon steels	1289–1478

	Medium carbon steels	1380–1514
	Low carbon steels	1480–1526
	Low alloy steels	1382–1529
	Stainless steels	1375–1450
Non-ferrous	Aluminum alloys	475–677
	Copper alloys	982–1082
	Lead alloys	322–328
	Magnesium alloys	447–649
	Nickel alloys	1435–1466
	Titanium alloys	1477–1682
	Zinc alloys	375–492
Glasses	Borosilicate glass (*)	450–602
	Glass ceramic (*)	563–1647
	Silica glass (*)	957–1557
	Soda-lime glass (*)	442–592
Ceramics		
Porous	Brick	927–1227
	Concrete, typical	927–1227
	Stone	1227–1427
Technical	Alumina	2004–2096
	Aluminum nitride	2397–2507
	Boron carbide	2372–2507
	Silicon	1407–1412
	Silicon carbide	2152–2500
	Silicon nitride	2388–2496
	Tungsten carbide	2827–2920
Composites		
Metal	Aluminum/silicon carbide	525–627
Polymer	CFRP	n/a
	GFRP	n/a
Natural	Bamboo (*)	77–102
	Cork (*)	77–102
	Leather (*)	107–127
	Wood, typical (longitudinal) (*)	77–102
	Wood, typical (transverse) (*)	77–102
Polymers		
Elastomer	Butyl Rubber (*)	–73––63
	EVA (*)	–73––23
	Isoprene (IR) (*)	–83––78
	Natural rubber (NR) (*)	–78––63
	Neoprene (CR) (*)	–48––43
	Polyurethane elastomers (elPU) (*)	–73––23
	Silicone elastomers (*)	–123––73
Thermoplastic	ABS (*)	88–128
	Cellulose polymers (CA) (*)	–9–107
	Ionomer (I) (*)	27–77
	Nylons (PA) (*)	44–56
	Polycarbonate (PC) (*)	142–205
	PEEK (*)	143–199
	Polyethylene (PE) (*)	–25––15
	PET (*)	68–80
	Acrylic (PMMA) (*)	85–165
	Acetal (POM) (*)	–18––8
	Polypropylene (PP) (*)	–25––15
	Polystyrene (PS) (*)	74–110
	Polyurethane thermoplastics (tpPU) (*)	120–160
	PVC	75–105
	Teflon (PTFE)	107–123
Thermoset	Epoxies	n/a
	Phenolics	n/a
	Polyester	n/a
Polymer foams		
	Flexible polymer foam (VLD) (*)	112–177
	Flexible polymer foam (LD) (*)	112–177

	Flexible Polymer foam (MD) (*)	112–177
	Rigid polymer foam (LD) (*)	67–171
	Rigid polymer foam (MD) (*)	67–157
	Rigid polymer foam (HD) (*)	67–171

1. The table lists the melting point for crystalline solids and the glass temperature for polymeric and inorganic glasses.

(*) glass transition temperature.

n/a: not applicable (materials decompose, rather than melt).

A5 Density, ρ

ρ (Mg/m ³)		
Metals		
Ferrous	Cast irons	7.05–7.25
	High carbon steels	7.8–7.9
	Medium carbon steels	7.8–7.9
	Low carbon steels	7.8–7.9
	Low alloy steels	7.8–7.9
	Stainless steels	7.6–8.1
Non-ferrous	Aluminum alloys	2.5–2.9
	Copper alloys	8.93–8.94
	Lead alloys	10–11.4
	Magnesium alloys	1.74–1.95
	Nickel alloys	8.83–8.95
	Titanium alloys	4.4–4.8
	Zinc alloys	4.95–7
Glasses	Borosilicate glass	2.2–2.3
	Glass ceramic	2.2–2.8
	Silica glass	2.17–2.22
	Soda-lime glass	2.44–2.49
Ceramics		
Porous	Brick	1.9–2.1
	Concrete, typical	2.2–2.6
	Stone	2.5–3
Technical	Alumina	3.5–3.98
	Aluminum nitride	3.26–3.33
	Boron carbide	2.35–2.55
	Silicon	2.3–2.35
	Silicon carbide	3–3.21
	Silicon nitride	3–3.29
	Tungsten Carbide	15.3–15.9
Composites		
Metal	Aluminum/Silicon Carbide	2.66–2.9
Polymer	CFRP	1.5–1.6
	GFRP	1.75–1.97
Natural	Bamboo	0.6–0.8
	Cork	0.12–0.24
	Leather	0.81–1.05
	Wood, typical (longitudinal)	0.6–0.8
	Wood, typical (transverse)	0.6–0.8
Polymers		
Elastomer	Butyl rubber	0.9–0.92
	EVA	0.945–0.955
	Isoprene (IR)	0.93–0.94
	Natural rubber (NR)	0.92–0.93
	Neoprene (CR)	1.23–1.25
	Polyurethane elastomers (elPU)	1.02–1.25
	Silicone elastomers	1.3–1.8
Thermoplastic	ABS	1.01–1.21
	Cellulose polymers (CA)	0.98–1.3
	Ionomer (I)	0.93–0.96
	Nylons (PA)	1.12–1.14
	Polycarbonate (PC)	1.14–1.21
	PEEK	1.3–1.32
	Polyethylene (PE)	0.939–0.96
	PET	1.29–1.4
	Acrylic (PMMA)	1.16–1.22

	Acetal (POM)	1.39–1.43
	Polypropylene (PP)	0.89–0.91
	Polystyrene (PS)	1.04–1.05
	Polyurethane thermoplastics (tpPU)	1.12–1.24
	PVC	1.3–1.58
	Teflon (PTFE)	2.14–2.2
Thermoset	Epoxies	1.11–1.4
	Phenolics	1.24–1.32
	Polyester	1.04–1.4
Polymer foams		
	Flexible polymer foam (VLD)	0.016–0.035
	Flexible polymer foam (LD)	0.038–0.07
	Flexible polymer foam (MD)	0.07–0.115
	Rigid polymer foam (LD)	0.036–0.07
	Rigid polymer foam (MD)	0.078–0.165
	Rigid Polymer foam (HD)	0.17–0.47

A6 Young's modulus, E

E (GPa)		
Metals		
Ferrous	Cast irons	165–180
	High carbon steels	200–215
	Medium carbon steels	200–216
	Low carbon steels	200–215
	Low alloy steels	201–217
	Stainless steels	189–210
Non-ferrous	Aluminum alloys	68–82
	Copper alloys	112–148
	Lead alloys	12.5–15
	Magnesium alloys	42–47
	Nickel alloys	190–220
	Titanium alloys	90–120
	Zinc alloys	68–95
Glasses	Borosilicate glass	61–64
	Glass ceramic	64–110
	Silica glass	68–74
	Soda-lime glass	68–72
Ceramics		
Porous	Brick	15–25
	Concrete, typical	25–38
	Stone	20–60
Technical	Alumina	215–413
	Aluminum nitride	302–348
	Boron carbide	400–472
	Silicon	140–155
	Silicon carbide	300–460
	Silicon nitride	280–310
	Tungsten carbide	600–720
Composites		
Metal	Aluminum/silicon carbide	81–100
Polymer	CFRP	69–150
	GFRP	15–28
Natural	Bamboo	15–20
	Cork	0.013–0.05
	Leather	0.1–0.5
	Wood, typical (longitudinal)	6–20
	Wood, typical (transverse)	0.5–3
Polymers		
Elastomer	Butyl rubber	0.001–0.002
	EVA	0.01–0.04
	Isoprene (IR)	0.0014–0.004
	Natural rubber (NR)	0.0015–0.0025
	Neoprene (CR)	0.0007–0.002
	Polyurethane elastomers (elPU)	0.002–0.003

	Silicone elastomers	0.005–0.02
Thermoplastic	ABS	1.1–2.9
	Cellulose polymers (CA)	1.6–2
	Ionomer (I)	0.2–0.424
	Nylons (PA)	2.62–3.2
	Polycarbonate (PC)	2–2.44
	PEEK	3.5–4.2
	Polyethylene (PE)	0.621–0.896
	PET	2.76–4.14
	Acrylic (PMMA)	2.24–3.8
	Acetal (POM)	2.5–5
	Polypropylene (PP)	0.896–1.55
	Polystyrene (PS)	2.28–3.34
	Polyurethane thermoplastics (tpPU)	1.31–2.07
	PVC	2.14–4.14
	Teflon (PTFE)	0.4–0.552
Thermoset	Epoxies	2.35–3.075
	Phenolics	2.76–4.83
	Polyester	2.07–4.41
Polymer foams		
	Flexible polymer foam (VLD)	0.0003–0.001
	Flexible polymer foam (LD)	0.001–0.003
	Flexible polymer foam (MD)	0.004–0.012
	Rigid polymer foam (LD)	0.023–0.08
	Rigid polymer foam (MD)	0.08–0.2
	Rigid polymer foam (HD)	0.2–0.48

A7 Yield strength, σ_y , and tensile strength, σ_{ts}

		σ_y (MPa)	σ_{ts} (MPa)
Metals			
Ferrous	Cast irons	215–790	350–1000
	High carbon steels	400–1155	550–1640
	Medium carbon steels	305–900	410–1200
	Low carbon steels	250–395	345–580
	Low alloy steels	400–1100	460–1200
	Stainless steels	170–1000	480–2240
Non-ferrous	Aluminum alloys	30–500	58–550
	Copper alloys	30–500	100–550
	Lead alloys	8–14	12–20
	Magnesium alloys	70–400	185–475
	Nickel alloys	70–1100	345–1200
	Titanium alloys	250–1245	300–1625
	Zinc Alloys	80–450	135–520
Glasses			
	Borosilicate glass (*)	264–384	22–32
	Glass ceramic (*)	750–2129	62–177
	Silica glass (*)	1100–1600	45–155
	Soda-lime glass (*)	360–420	31–35
Ceramics			
Porous	Brick (*)	50–140	7–14
	Concrete, typical (*)	32–60	2–6
	Stone (*)	34–248	5–17
Technical	Alumina (*)	690–5500	350–665
	Aluminum nitride (*)	1970–2700	197–270
	Boron carbide (*)	2583–5687	350–560
	Silicon (*)	3200–3460	160–180
	Silicon carbide (*)	1000–5250	370–680
	Silicon nitride (*)	524–5500	690–800
	Tungsten carbide (*)	3347–6833	370–550
Composites			
Metal	Aluminum/silicon carbide	280–324	290–365
Polymer	CFRP	550–1050	550–1050
	GFRP	110–192	138–241
Natural	Bamboo	35–44	36–45

	Cork	0.3–1.5	0.5–2.5
	Leather	5–10	20–26
	Wood, typical (longitudinal)	30–70	60–100
	Wood, typical (transverse)	2–6	4–9
Polymers			
Elastomer	Butyl rubber	2–3	5–10
	EVA	12–18	16–20
	Isoprene (IR)	20–25	20–25
	Natural rubber (NR)	20–30	22–32
	Neoprene (CR)	3.4–24	3.4–24
	Polyurethane elastomers (elPU)	25–51	25–51
	Silicone elastomers	2.4–5.5	2.4–5.5
Thermoplastic	ABS	18.5–51	27.6–55.2
	Cellulose polymers (CA)	25–45	25–50
	Ionomer (I)	8.3–15.9	17.2–37.2
	Nylons (PA)	50–94.8	90–165
	Polycarbonate (PC)	59–70	60–72.4
	PEEK	65–95	70–103
	Polyethylene (PE)	17.9–29	20.7–44.8
	PET	56.5–62.3	48.3–72.4
	Acrylic (PMMA)	53.8–72.4	48.3–79.6
	Acetal (POM)	48.6–72.4	60–89.6
	Polypropylene (PP)	20.7–37.2	27.6–41.4
	Polystyrene (PS)	28.7–56.2	35.9–56.5
	Polyurethane thermoplastics (tpPU)	40–53.8	31–62
	PVC	35.4–52.1	40.7–65.1
	Teflon (PTFE)	15–25	20–30
Thermoset	Epoxies	36–71.7	45–89.6
	Phenolics	27.6–49.7	34.5–62.1
	Polyester	33–40	41.4–89.6
Polymer foams			
	Flexible polymer foam (VLD)	0.01–0.12	0.24–0.85
	Flexible polymer foam (LD)	0.02–0.3	0.24–2.35
	Flexible polymer foam (MD)	0.05–0.7	0.43–2.95
	Rigid polymer foam (LD)	0.3–1.7	0.45–2.25
	Rigid polymer foam (MD)	0.4–3.5	0.65–5.1
	Rigid polymer foam (HD)	0.8–12	1.2–12.4

(*) NB: For ceramics, yield strength is replaced by *compressive strength*, which is more relevant in ceramic design. Note that ceramics are of the order of 10 times stronger in compression than in tension.

A8 Fracture toughness (plane-strain), K_{Ic}

K_{Ic} (MPa√m)		
Metals		
Ferrous	Cast irons	22–54
	High carbon steels	27–92
	Medium carbon steels	12–92
	Low carbon steels	41–82
	Low alloy steels	14–200
	Stainless steels	62–280
Non-ferrous	Aluminum alloys	22–35
	Copper alloys	30–90
	Lead alloys	5–15
	Magnesium alloys	12–18
	Nickel alloys	80–110
	Titanium alloys	14–120
	Zinc alloys	10–100
Glasses	Borosilicate glass	0.5–0.7
	Glass ceramic	1.4–1.7
	Silica glass	0.6–0.8
	Soda-lime glass	0.55–0.7
Ceramics		
Porous	Brick	1–2
	Concrete, typical	0.35–0.45
	Stone	0.7–1.5
Technical	Alumina	3.3–4.8

	Aluminum nitride	2.5–3.4
	Boron carbide	2.5–3.5
	Silicon	0.83–0.94
	Silicon carbide	2.5–5
	Silicon nitride	4–6
	Tungsten carbide	2–3.8
Composites		
Metal	Aluminum/silicon carbide	15–24
Polymer	CFRP	6.1–88
	GFRP	7–23
Natural	Bamboo	5–7
	Cork	0.05–0.1
	Leather	3–5
	Wood, typical (longitudinal)	5–9
	Wood, typical (transverse)	0.5–0.8
Polymers		
Elastomer	Butyl rubber	0.07–0.1
	EVA	0.5–0.7
	Isoprene (IR)	0.07–0.1
	Natural rubber (NR)	0.15–0.25
	Neoprene (CR)	0.1–0.3
	Polyurethane elastomers (elPU)	0.2–0.4
	Silicone elastomers	0.03–0.5
Thermoplastic	ABS	1.19–4.30
	Cellulose polymers (CA)	1–2.5
	Ionomer (I)	1.14–3.43
	Nylons (PA)	2.22–5.62
	Polycarbonate (PC)	2.1–4.60
	PEEK	2.73–4.30
	Polyethylene (PE)	1.44–1.72
	PET	4.5–5.5
	Acrylic (PMMA)	0.7–1.6
	Acetal (POM)	1.71–4.2
	Polypropylene (PP)	3–4.5
	Polystyrene (PS)	0.7–1.1
	Polyurethane thermoplastics (tpPU)	1.84–4.97
	PVC	1.46–5.12
	Teflon (PTFE)	1.32–1.8
Thermoset	Epoxies	0.4–2.22
	Phenolics	0.79–1.21
	Polyester	1.09–1.70
Polymer foams		
	Flexible polymer foam (VLD)	0.005–0.02
	Flexible polymer foam (LD)	0.015–0.05
	Flexible polymer foam (MD)	0.03–0.09
	Rigid polymer foam (LD)	0.002–0.02
	Rigid polymer foam (MD)	0.007–0.049
	Rigid polymer foam (HD)	0.024–0.091

Note: K_{IC} is only valid for conditions under which linear elastic fracture mechanics apply (see Chapter 8). The plane-strain toughness, G_{IC} , may be estimated from $K_{IC}^2 = E G_{IC} / (1 - \nu^2) \approx E G_{IC}$ (as $\nu^2 \approx 0.1$).

A9 Thermal conductivity, λ , and thermal expansion, α

		λ (W/m.K)	α (10 ⁻⁶ /C)
Metals			
Ferrous	Cast irons	29–44	10–12.5
	High carbon steels	47–53	11–13.5
	Medium carbon steels	45–55	10–14
	Low carbon steels	49–54	11.5–13
	Low alloy steels	34–55	10.5–13.5
	Stainless steels	11–19	13–20
Non-ferrous	Aluminum alloys	76–235	21–24
	Copper alloys	160–390	16.9–18
	Lead alloys	22–36	18–32
	Magnesium alloys	50–156	24.6–28

	Nickel alloys	67-91	12-13.5
	Titanium alloys	5-12	7.9-11
	Tungsten alloys	100-142	4-5.6
	Zinc alloys	100-135	23-28
Glasses	Borosilicate glass	1-1.3	3.2-4.0
	Glass ceramic	1.3-2.5	1-5
	Silica glass	1.4-1.5	0.55-0.75
	Soda-lime glass	0.7-1.3	9.1-9.5
Ceramics			
Porous	Brick	0.46-0.73	5-8
	Concrete, typical	0.8-2.4	6-13
	Stone	5.4-6.0	3.7-6.3
Technical	Alumina	30-38.5	7-10.9
	Aluminum nitride	80-200	4.9-6.2
	Boron carbide	40-90	3.2-3.4
	Silicon	140-150	2.2-2.7
	Silicon carbide	115-200	4.0-5.1
	Silicon nitride	22-30	3.2-3.6
	Tungsten carbide	55-88	5.2-7.1
Composites			
Metal	Aluminum/silicon carbide	180-160	15-23
Polymer	CFRP	1.28-2.6	1-4
	GFRP	0.4-0.55	8.6-33
Natural	Bamboo	0.1-0.18	2.6-10
	Cork	0.035-0.048	130-230
	Leather	0.15-0.17	40-50
	Wood, typical (longitudinal)	0.31-0.38	2-11
	Wood, typical (transverse)	0.15-0.19	32-42
Polymers			
Elastomer	Butyl rubber	0.08-0.1	120-300
	EVA	0.3-0.4	160-190
	Isoprene (IR)	0.08-0.14	150-450
	Natural rubber (NR)	0.1-0.14	150-450
	Neoprene (CR)	0.08-0.14	575-610
	Polyurethane elastomers	0.28-0.3	150-165
	Silicone elastomers	0.3-1.0	250-300
Thermoplastic	ABS	0.19-0.34	84.6-234
	Cellulose polymers (ca)	0.13-0.3	150-300
	Ionomer (I)	0.24-0.28	180-306
	Nylons (PA)	0.23-0.25	144-150
	Polycarbonate (PC)	0.19-0.22	120-137
	PEEK	0.24-0.26	72-194
	Polyethylene (PE)	0.40-0.44	126-198
	PET	0.14-0.15	114-120
	Acrylic (PMMA)	0.08-0.25	72-162
	Acetal (POM)	0.22-0.35	76-201
	Polypropylene (PP)	0.11-0.17	122-180
	Polystyrene (PS)	0.12-0.12	90-153
	Polyurethane thermoplastics	0.23-0.24	90-144
	PVC	0.15-0.29	100-150
	Teflon (PTFE)	0.24-0.26	126-216
Thermoset	Epoxies	0.18-0.5	58-117
	Phenolics	0.14-0.15	120-125
	Polyester	0.28-0.3	99-180
Polymer foams			
	Flexible polymer foam (VLD)	0.036-0.048	120-220
	Flexible polymer foam (LD)	0.04-0.06	115-220
	Flexible polymer foam (MD)	0.04-0.08	115-220
	Rigid polymer foam (LD)	0.023-0.04	20-80
	Rigid polymer foam (MD)	0.027-0.038	20-75
	Rigid polymer foam (HD)	0.034-0.06	22-70

A10 Electrical resistivity and dielectric constant

Resistivity ($\mu\text{ohm.cm}$)		Dielectric constant	

Metals			
Ferrous	Cast irons	49–56	–
	High carbon steels	17–20	–
	Medium carbon steels	15–22	–
	Low carbon steels	15–20	–
	Low alloy steels	15–35	–
	Stainless steels	64–107	–
Non-ferrous	Aluminum alloys	2.5–5.0	–
	Copper alloys	1.7–24	–
	Lead alloys	15–22	–
	Magnesium alloys	4.2–15	–
	Nickel alloys	6–114	–
	Titanium alloys	100–170	–
	Tungsten alloys	10.2–14	–
	Zinc alloys	5.4–7.2	–
Glasses	Borosilicate glass	3×10^{21} – 3×10^{22}	4.6–6.0
	Glass ceramic	2×10^{19} – 1×10^{21}	5.3–6.2
	Silica glass	1×10^{23} – 1×10^{27}	3.7–3.9
	Soda-lime glass	8×10^{17} – 8×10^{18}	7.0–7.6
Ceramics			
Porous	Brick	1×10^{14} – 3×10^{16}	7.0–10
	Concrete, typical	1.8×10^{12} – 1.8×10^{13}	8.0–12
	Stone	1×10^8 – 1×10^{14}	6.0–18
Technical	Alumina	1×10^{20} – 1×10^{22}	6.5–6.8
	Aluminum nitride	1×10^{19} – 1×10^{21}	8.3–9.3
	Boron carbide	1×10^5 – 1×10^7	4.8–8.0
	Silicon	1×10^6 – 1×10^{12}	11–12
	Silicon carbide	1×10^9 – 1×10^{12}	6.3–9.0
	Silicon nitride	1×10^{20} – 1×10^{21}	7.9–8.1
	Tungsten carbide	20–100	–
Composites			
Metal	Aluminum/silicon carbide	5–12	–
Polymer	CFRP	1.7×10^5 – 1×10^6	–
	GFRP	1×10^{16} – 2×10^{22}	4.2–5.2
Natural	Bamboo	6×10^{13} – 7×10^{14}	5–7
	Cork	1×10^9 – 1×10^{11}	6–8
	Leather	1×10^8 – 1×10^{10}	5–10
	Wood, typical (longitudinal)	6×10^{13} – 2×10^{14}	5–6
	Wood, typical (transverse)	2×10^{14} – 7×10^{14}	5–6
Polymers			
Elastomer	Butyl rubber	1×10^{15} – 1×10^{16}	2.8–3.2
	EVA	3.2×10^{21} – 1×10^{22}	2.9–3.0
	Isoprene (IR)	1×10^{15} – 1×10^{16}	2.5–3.0
	Natural rubber (NR)	1×10^{15} – 1×10^{16}	3.0–4.5
	Neoprene (CR)	1×10^{19} – 1×10^{23}	6.7–8.0
	Polyurethane elastomers	1×10^{18} – 1×10^{22}	5.0–9.0
	Silicone elastomers	3×10^{19} – 1×10^{22}	2.9–4.0
Thermoplastic	ABS	3×10^{21} – 3×10^{22}	2.8–3.2
	Cellulose polymers (ca)	1×10^{17} – 4×10^{20}	3.0–5.0
	Ionomer (I)	3×10^{21} – 3×10^{22}	2.2–2.4
	Nylons (PA)	1.5×10^{19} – 1.1×10^{20}	3.7–3.9
	Polycarbonate (PC)	1×10^{20} – 1×10^{21}	3.1–3.3
	PEEK	3×10^{21} – 3×10^{22}	3.1–3.3
	Polyethylene (PE)	3×10^{22} – 3×10^{24}	2.2–2.4
	PET	3×10^{20} – 3×10^{21}	3.5–3.7
	Acrylic (PMMA)	3×10^{23} – 3×10^{24}	3.2–3.4
	Acetal (POM)	3×10^{20} – 3×10^{21}	3.6–4.0
	Polypropylene (PP)	3×10^{22} – 3×10^{23}	2.1–2.3
	Polystyrene (PS)	1×10^{25} – 1×10^{27}	3.0–3.3
	Polyurethane thermoplastics	3×10^{18} – 3×10^{19}	6.6–7.1
	PVC	1×10^{20} – 1×10^{22}	3.1–4.4
	Teflon (PTFE)	3×10^{23} – 3×10^{24}	2.1–2.2
Thermoset	Epoxies	1×10^{20} – 6×10^{21}	3.4–5.7

	Phenolics	3×10^{18} – 3×10^{19}	4.0–6.0
	Polyester	3×10^{18} – 3×10^{19}	2.8–3.3
Polymer foams			
	Flexible polymer foam (VLD)	1×10^{20} – 1×10^{23}	1.1–1.15
	Flexible polymer foam (LD)	1×10^{20} – 1×10^{23}	1.15–1.2
	Flexible polymer foam (MD)	1×10^{20} – 1×10^{23}	1.2–1.3
	Rigid polymer foam (LD)	1×10^{17} – 1×10^{21}	1.04–1.1
	Rigid polymer foam (MD)	3×10^{16} – 3×10^{20}	1.1–1.19
	Rigid polymer foam (HD)	1×10^{16} – 1×10^{20}	1.2–1.45